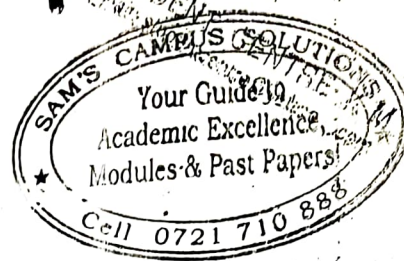


ECU

109



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2015/2016
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE
ECU 109: FUNDAMENTALS OF COMPUTING

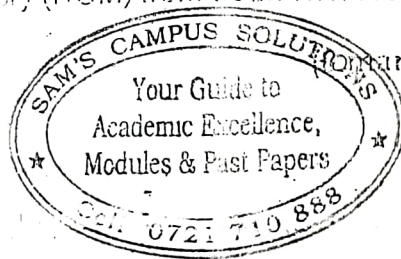
DATE: FRIDAY 11TH DECEMBER 2015 TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS:

Answer Question ONE and any TWO questions

Question One

- (a) Based on the definition of a computer, list the four major components of a computer. (4 marks)
- (b) Outline development Read Only Memory (ROM) from traditional ROM through to Electronically EPROM. (4 marks)
- (c) Carry out the following calculations. (10 marks)
- (i) $110010_2 + 101101_2 + 110011_2$
 - (ii) $546_8 + 272_8$
 - (iii) $FAC_{16} - EED_{16}$
 - (iv) $327_8 + FC9_{16}$
- (d) As a computer network expert give two reasons that would make you strongly advice your organization to develop local area network. (4 marks)
- (e) Differentiate between an interpreter and a compiler. (4 marks)



Question Two

- (a) Describe any three characteristics of the fifth generation that makes it different from all the other generations of computer. (6 marks)
- (b) Outline the three points behind the store program concept suggested by John Von Neumann. (6 marks)

- (c) Describe the magnetic ink character reader (MICR) and give its area of application. (3 marks)

Question Three

- (a) Describe any three editing features of a word processor. (6 marks)
- (b) Outline the steps you would follow to perform the following in a word processor.
- (i) mailmerge
 - (ii) insert a picture
 - (iii) subscript an existing text
- (9 marks)

Question Four

- (a) In excel one can create a formula where the same cell can always be referred whenever a formula is created. Use a suitable example to demonstrate how this can be done. (6 marks)
- (b) Suppose you are a bread supplier and you want to keep track of the payment of your customers and final amount of money collected at the end of the day's sales. You have decided to use the spreadsheet program to create a table to help you do that. Below is a sample table.

	A	B	C	D	E
1	Client's Name	No. of crates	Amount	Unit Price	
2	Ndutu superMarket	275	(a)	120	
3	Osongore shop	18	(b)		
4	Jipe Moyo Minishop	5	(c)		
5	Umoja shop	90	(d)		
6	Total Collection		(e)		
7					

Required:

Create a formula to calculate the value at (a). Using the copy feature get the amount for (b) – (d). Create a formula to calculate the total collection.

(9 marks)

Question Five

- (a) Define the following terms:
- (i) Concentric computing
 - (ii) Cloud computing



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)
ECU 109: FUNDAMENTAL OF COMPUTING

DATE: Friday, 23rd February 2018

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS: Answer question ONE and any TWO questions.

Question One

- (a) Define the term computer. (2 marks)
- (b) Explain role of the buffers found in the input/output module. (4 marks)
- (c) With aid of suitable diagrams explain the three basic configurations of computer network (9 marks)
- (d) Most organizations have embraced computer technology. Explain any three benefits that your organization will get by using computers in their operations. (6 marks)
- (e) Carry out the following computations: (9 marks)
 - (i) $10101_2 + 11111_2 + 11001_2$
 - (ii) $F954_{16} - 6AE9_{16}$
 - (iii) $FC8_{16} - 7432_8$

Question Two

- (a) Distinguish between the RAM and ROM of the main memory. (6 marks)
- (b) With aid of suitable diagram discuss the memory hierarchy. (9 marks)

Question Three

- (a) Briefly explain the steps one would follow to do the following:
 - (i) Insert footer into a given document.
 - (ii) To Italic an existing block of text.

- (iii) To page number a document (9 marks)
- (b) Discuss how you will insert the following WordArt "PLEASE COME HOME" in an existing document. (6 marks)

Question Four

- (a) Give two reasons that explain why an electronic spreadsheet is better than a manual spreadsheet. (4 marks)
- (b) The table below shows the marks obtained in given class in one of the Nairobi County.

	A	B	C	D	E	F
1	S/No.	Customer Name	No. crates	Amount		Unit Price
2	1	Kassmart Supermarket	334			225
3	2	Wisdom mini-shop	48			
4	3	Umoja Shop	85			
5	4	Osongore Milkshop	540			
6	5	Anne's Shop	34			
7	6	Green Valley shop	75			
8						
9		Total Collection				
10		Average Collection				
11						
12						

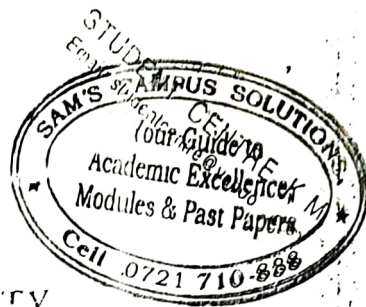
Required:

- Create a formula to calculate the amount paid by Kassmart supermarket.
- Using the copy feature get the amount paid by each customer for the supplies made.
- Give the formula to calculate total collection.
- Give the formula for calculating the average collection.

(11 marks)

Question Five

- (a) Explain why security in computer systems has become something of concern. (3 marks)
- (b) The National Computing Centre specifies four principles of Security. Discuss these four principles. (12 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)

ECU 109: FUNDAMENTALS OF COMPUTING

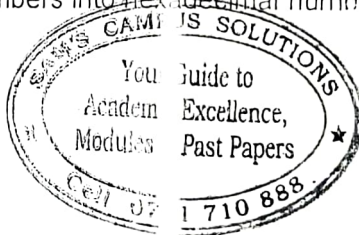
DATE: Friday, 9th December 2016

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS: Answer question ONE and any TWO questions.

Question One

- (a) Central processing unit has three internal functions components, with the aid of suitable diagram explain how these components communicate with each other. (5 marks)
- (b) Briefly explain why the Input/output devices are not connected directly to the central processing unit. (2 marks)
- (c) Explain why an institution can opt to develop a local area network. (4 marks)
- (d) List any five security threats your computer may face. (5 marks)
- (e) List the five ~~three~~ generations of computer languages. (5 marks)
- (f) Convert the following numbers into hexadecimal numbering system:
 - (i) 1904_{10}
 - (ii) 1111101001110_2
 - (iii) 7412_8



(9 marks)

Question Two

- (a) With aid of suitable diagram discuss the memory hierarchy. (9 marks)
- (b) Discuss the two factors behind the development of the Input devices giving an example for each factor. (6 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Question Three

- (a) Using simple illustrations, explain the two main functions of the operating system. (6 marks)
- (b) Explain any three resources managed by the operating system clearly showing how the operating system manages the resource. (9 marks)

Question Four

- (a) Using suitable examples, explain any three types of data supported by a spreadsheet program. (6 marks)
- (b) The table below shows the marks obtained in given class in one of the Nairobi County.

S.No.	Name	Maths	English	Kiswahili	CRE	Science	Total
	Peter Njau	55	85	80	72	78	
	Alice Jerotich	78	89	57	60	88	
	Amos Owino	85	90	78	62	40	
	John Ochiave	50	75	90	64	59	
	Jane Wanjiru	45	59	86	88	76	
	Joel Wanjala	60	77	50	80	74	
	Noah Ngeno	49	89	45	76	55	
	Ann Aliene	57	46	49	66	60	
	Joyce Njoroge	66	54	70	60	76	

Required:

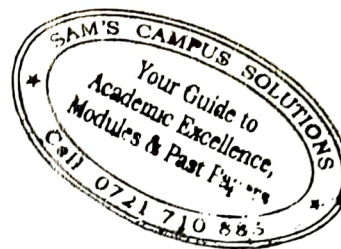
- (i) Create a formula to calculate the value the total for the first student "Peter Njau"
- (ii) Using the copy feature get the totals for the rest of the students
- (iii) Provide the steps that you will use to sort the students based on the total column from the highest mark to the lowest mark. (9 marks)

Question Five

- (a) Discuss any three formatting features of a word processing program giving an example for each. (9 marks)
- (b) Discuss the mail merge feature in word processing and how you can apply it in our learning institution. (6 marks)

STUDENT CENTRE
E-mail: studentcentre@...
www...

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KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2014/2015
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)

ECU 109: FUNDAMENTALS OF COMPUTING

DATE: Wednesday, 10th December, 2014

TIME: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS: Answer question ONE and any TWO questions.

Question One

- (a) Define the following terms:
(i) Hardcopy
(ii) Byte
(iii) microfilm
(iv) Utility program
(4 marks)
- (b) Differentiate between a compiler and an interpreter.
(2 marks)
- (c) Briefly explain the first four generations of computer clearly stating the technology used in each of the generation.
(8 marks)
- (d) Briefly explain the four basic resources that are managed by any operating system.
(8 marks)
- (e) Carry out the following arithmetic calculations:
(i) $FAD3_{16} + 5BCE_{16}$
(ii) $1111100_2 + 1101001_2 + 1011011_2$
(iii) $4D8_{16} + 724_8$ giving your answer in binary
(8 marks)

Question Two

Briefly write the procedures of carrying the following tasks in wordprocessing:

- (i) Indenting a paragraph from both left and right margin.
(ii) Italics and underlining an existing text
(iii) Copying a block of text between two documents

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 1 of 2

- (iv) Inserting a row into an existing table.
(v) Spell checking a document



Question Three

- (a) State and explain the four data types that are supported by a spreadsheet.

(8 marks)

- (b) Use the table below to answer the following questions:

	A	B	C	D	E	F	G
1	Item/month	Jan	Feb	March	April	May	June
2	Food	2500	3000	1750	5200	4200	5000
3	Clothing	2000	1500	4000	960	350	1200
4	Electricity Bill	450	700	1250	600	1000	860
5	Water Bill	146	200	190	340	700	250
6	Total Expenditure				(a)		
7	Average Expenditure	(b)					
8	Maximum Value	(c)					

Required:

- (i) Create a formula to calculate the value at (a), (b) and (c).
(ii) Briefly explain how you can represent water bill variable on a pie chart.

(12 marks)

Question Four

- (a) State and explain any five threats to a computer system
(b) List and explain any five measures that can be put in place to minimize the risk of unauthorized data access and fraud.

(10 marks)

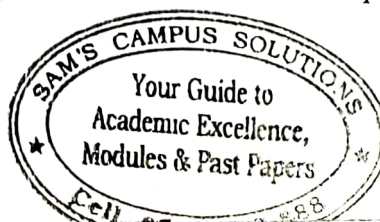
Question Five

- (a) With aid of a suitable diagram explain the basic components of a computer network.
(b) State and explain the three network topologies.
(c) Explain any three benefits that your institution can get by using the Internet

(6 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 2 of 2





KENYATTA UNIVERSITY
ELECTRICAL AND ELECTRONICS DEPARTMENT

ECU109: FUNDAMENTALS OF COMPUTING

CAT1

14th January, 2021

TIME: 1 HOUR

INSTRUCTIONS: Answer all questions

(a) Define the following terms:

- (i) RAM
- (ii) Control Unit
- (iii) Data
- (iv) Bit

(4 marks)

(b) Carry out the following arithmetic calculations:

- (i) $F9A4_{16} + D7E5_{16}$ *512 256 128 64 32 16 8 4 2 1*
- (ii) $1101011_2 + 1011111_2 + 1111111_2$ *1 7 7 655*
- (iii) $CF6_{16} + 362_8$

(8 marks)

(c) Using a suitable diagram discuss the computer memory hierarchy.

(10 marks)

(d) With aid of a suitable diagram explain the functional parts of the computer.

cache

(8 marks)

10 11 12 13 14 15

A B C D E F

3

2



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

**FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)**

ECU 109: FUNDAMENTALS OF COMPUTING

DATE: Thursday, 11th February 2020

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS: Answer Question ONE and any other TWO Questions

Question One

- (a) Define the term buffer as used in computer field. (2 marks)
- (b) The Central Processing Unit has three functional components; with the aid of a suitable diagram explain these three components. (8 marks)
- (c) Operating systems manages the computer resource, State the four activities involved in the management of these resources. (4 marks)
- (d) When securing any computer system there three objectives to be attained, explain these three objectives. (6 marks)
- (e) Carry out the following calculations.
 - (i) $11111011_2 + 10111111_2 + 11011101_2$
 - (ii) $E790_{16} - 2CD8_{16}$
 - (iii) $F1A3_{16} + 6037_8$(10 marks)

Question Two

- (a) With aid of a suitable diagram discuss the three basic hardware components of data communication network. (9 marks)
- (b) Most organization are now adopting Local Area Network. Discuss two reasons that you may use to convince your organization to adopt a Local Area Network. (6 marks)

Question Three

Assuming you are the Secretary General for the nyumba kumi community. The management has decided to employ staff to run the secretariat of the organization. The organization has advertised the positions and received the applications. The management has instructed you to send invitation letters to the applicants to attend interviews that will be spread over four days, three panels and each applicant has been allocated day, panel and time slot. Discuss how you will use the mail merge feature in word processing to accomplish this task. (15 marks)

Question Four

- (a) Using simple illustration discuss the benefits of using a cell reference concept as opposed to actual values. (6 marks)
- (b) You have been contracted to sell bread to various customers by the superloaf bakery. As a supplier you have decided to create a spreadsheet of your customers to help you keep records of your sales. The table below shows part of the data you are keeping that will help you keep track of your collections on a daily basis.

	A	B	C	D	E	F
1	S/No	Customer Name	No. crates	Amount		Unit Price
2	1	Majengo Minishop	1314			460
3	2	Huruma Shop	708			
4	3	Kiamumbi Supermarket	97			
5	4	Mbuzi Bread	340			
6	5	Magge's Shop	39			
7	6	Black Cottage shop	185			
8						
9		Total Collection				
10		Average Collection				
11						
12						

Required:

- (i) Create a formula to calculate the amount paid by Majengo Minishop. (2 marks)
- (ii) Using the copy feature illustrate how you will get the amount paid by each customer for the supplies made. (5 marks)
- (iii) Create the formula to calculate total amount collected. (1 mark)

- (iv) Create the formula to calculate the average amount collected.
(1 mark)

QUESTION FIVE (20 MARKS)

- (a) With aid of a suitable diagram discuss the functions of the Input/output module.
(6 marks)
- (b) Discuss the three lines that constitute the bus systems.
(9 marks)